

How Much Are We Underestimating Global Coral Bleaching Severity?

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Background

- Coral reefs are highly vulnerable to climate change
- Bleaching driven by heat stress (DHW)
- Global coral bleaching databases rely on **field surveys**
- Surveys often miss **peak heat stress timing**

→ Result: potential **underestimation in global coral bleaching severity**

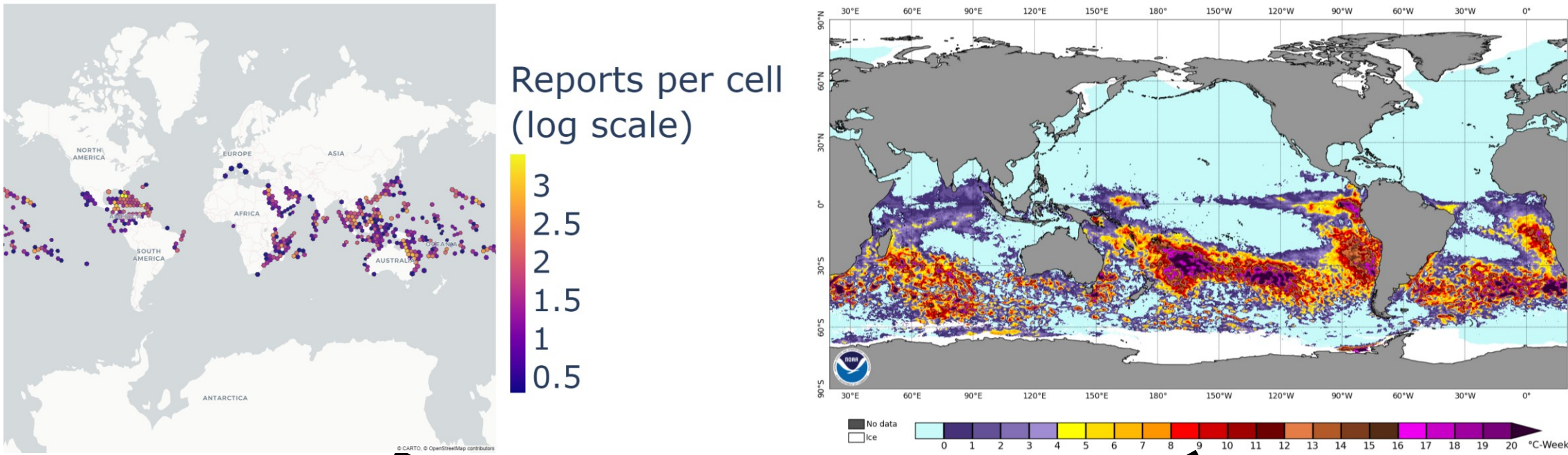
Research Question

How much is global coral bleaching severity underestimate due to survey timing?

Methods – Data Assembly

Global Coral Bleaching Database

Daily Global 5km Satellite Coral Bleaching Heat Stress metric (DHW)



Expanded Global Coral Bleaching Database

- DHW at time of survey for each bleaching report

+/- 6 month rolling window

Expanded Global Coral Bleaching Database

- Maximum DHW within rolling window: DHW_{max}
- Date of DHW_{max}
- Time lag between survey date and date of DHW_{max} : t_{lag}

Final Global Coral Bleaching Database

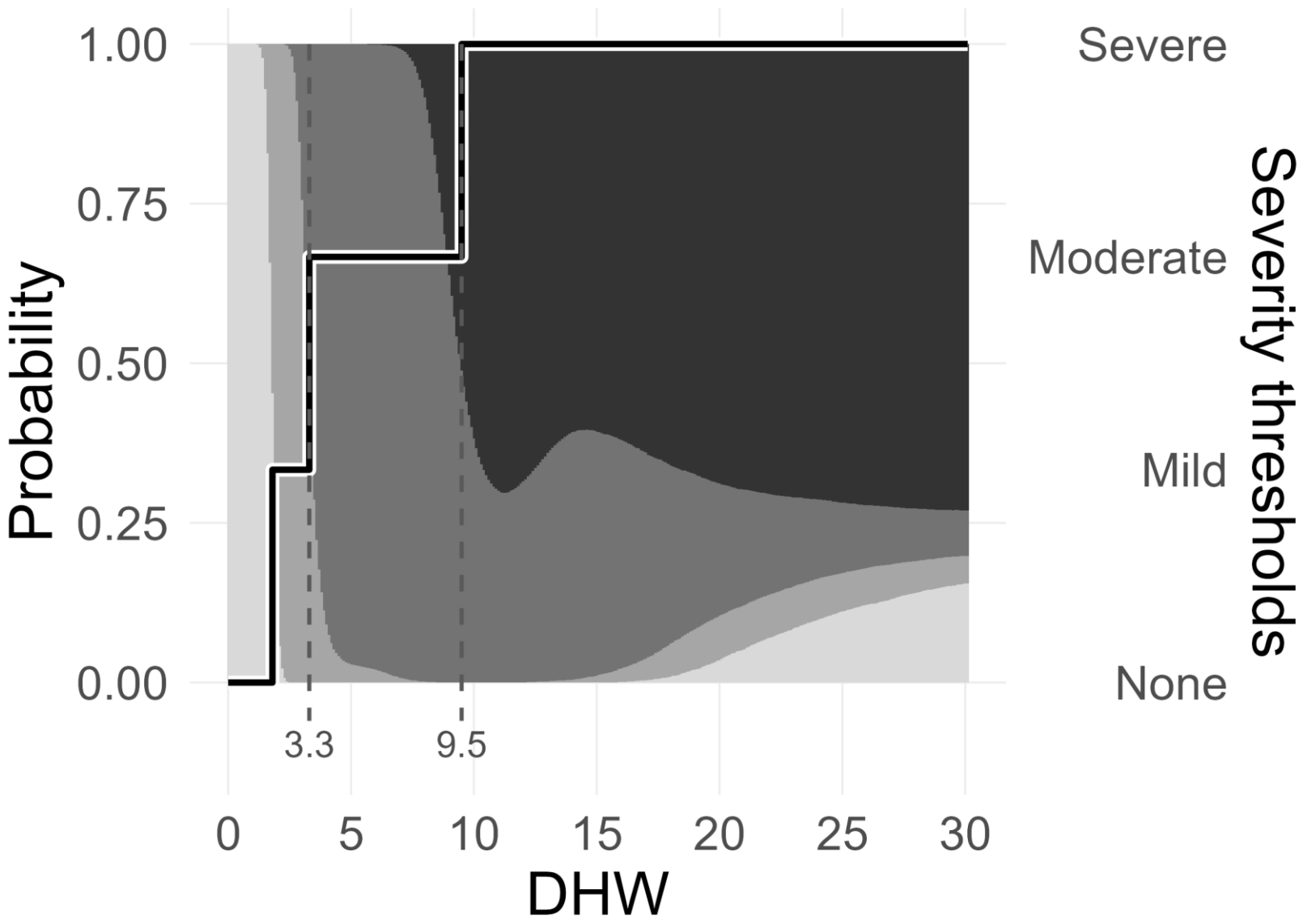
- Expanded database with predictor and response variables
- ~16,000 bleaching reports for modeling purposes

Methods – Modelling Approach

- Two Bayesian ordinal models include fixed effects, and spatial and spatio-temporal random effects
- They capture bleaching severity before and after peak heat stress

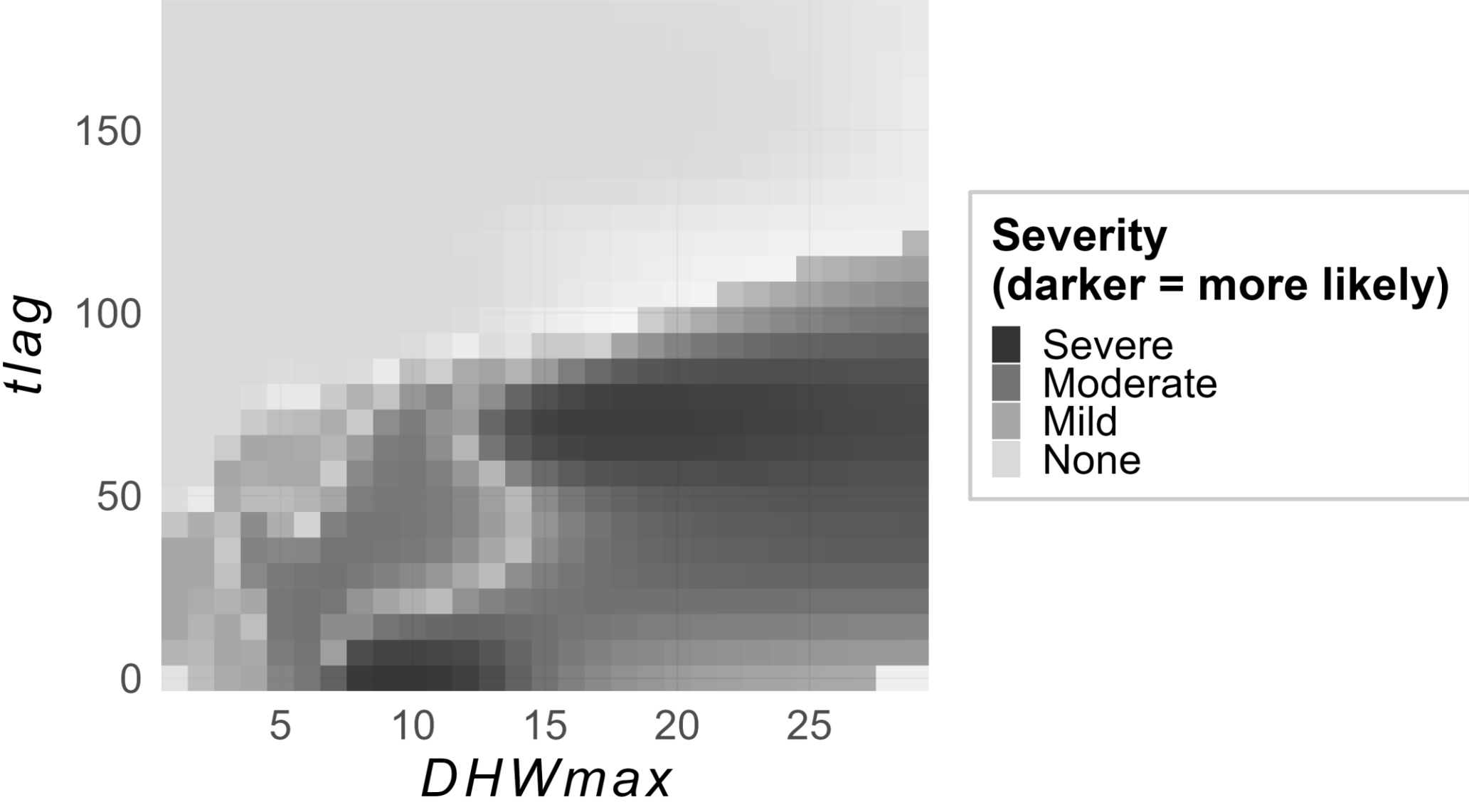
Pre-peak model (before maximum heat stress)

- Severity increases with accumulated DHW
- Captures *onset* and *intensification* of bleaching

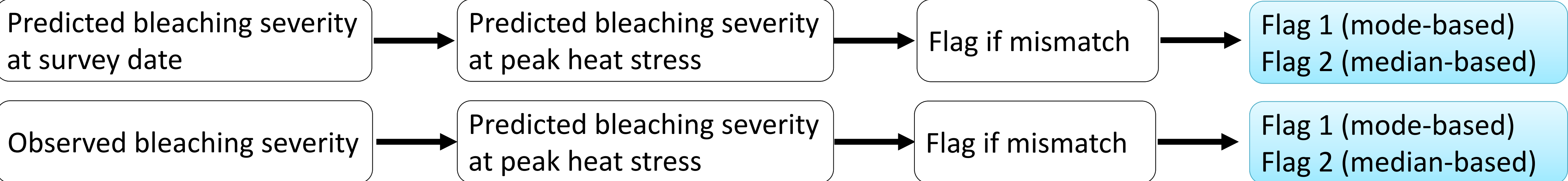


Post-peak model (after maximum heat stress)

- Severity depends on DHW_{max} and t_{lag}
- Captures *recovery dynamics*



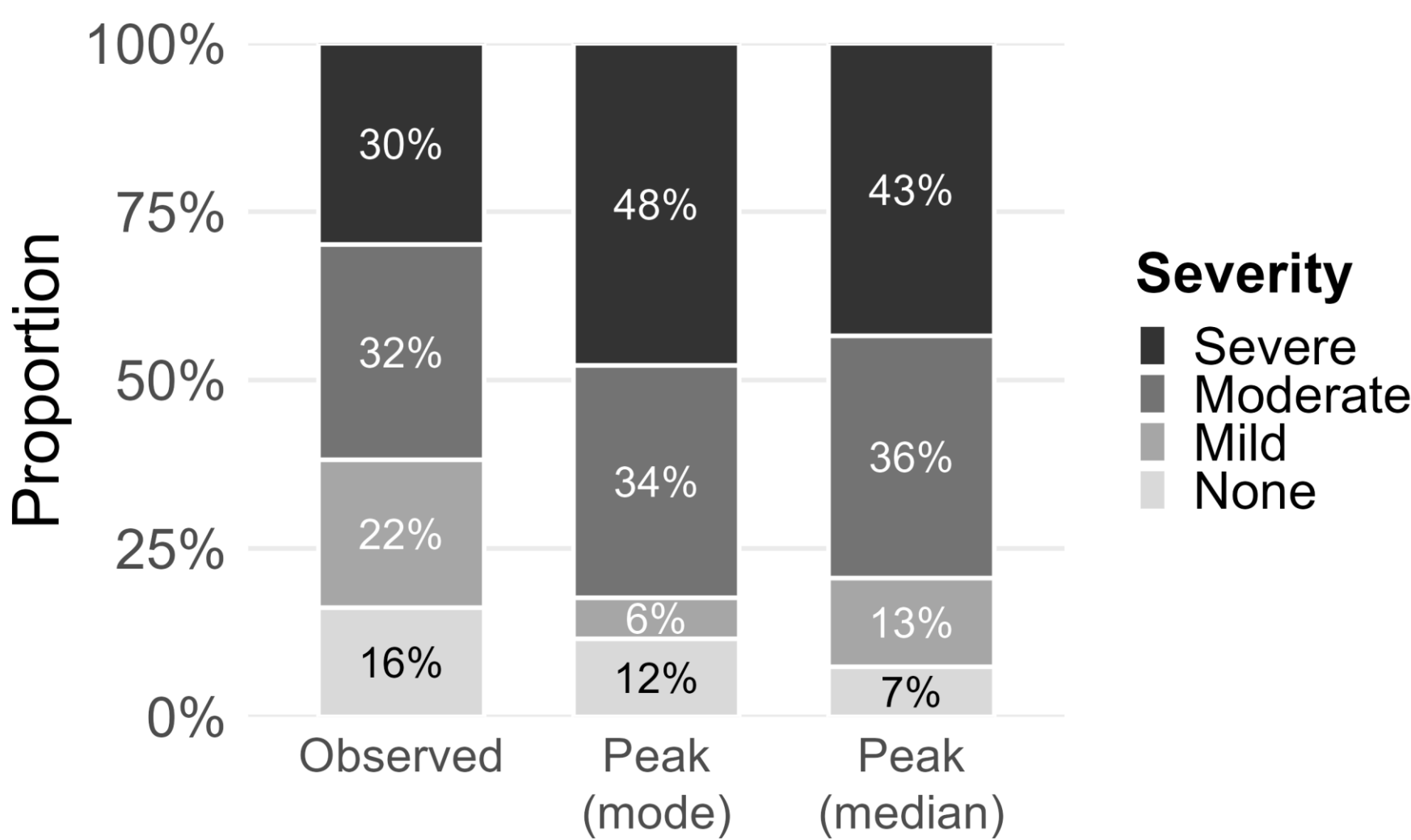
Methods – Detection of survey timing bias



Key Results

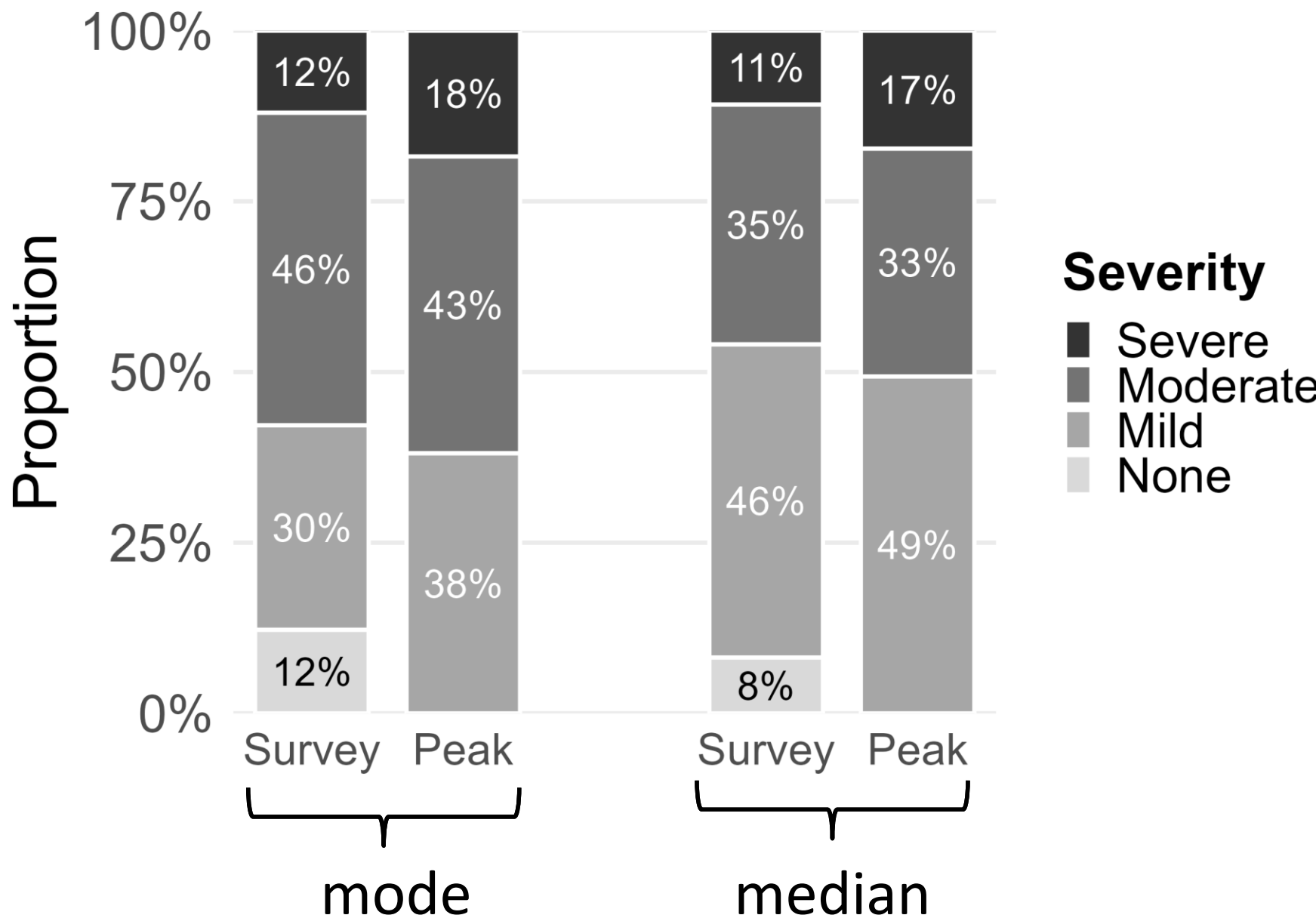
Pre-peak model

- 58% flagged at least once
- 16% flagged under all 4 rules
- Severe bleaching likely underestimated globally by 13%-18%



Post-peak model

- 45% flagged at least once
- 9% flagged under all 4 rules
- Severe bleaching likely underestimated globally by 6% due to survey timing



Implications

- ❑ **Global bleaching is systematically underestimated**
 - Missed peak conditions → lower reported severity
 - Extreme bleaching events likely underreported
- ❑ **Bias in heat stress-bleaching relationships**
 - Up to 45% of post-peak reports flagged as underestimated
 - Post-peak surveys capture recovery, not peak conditions
 - This decouples bleaching severity from DHW
- ❑ **Consequences for models and projections**
 - Inflated thermal thresholds
 - Weakened statistical relationships
 - Reduced model performance and interpretability.

Recommendations

- ❑ **Use global coral bleaching databases with caution**
 - Account for timing bias in analyses
- ❑ **Align surveys with peak heat stress wherever possible**
 - NOAA Regional Virtual Stations shows when DHW peaked in previous years
 - NOAA four-month bleaching outlooks
 - NOAA satellite bleaching alert systems

Novelty

- ❑ **Probabilistic framework** for global bleaching patterns
- ❑ Spatio-temporal hierarchical structure capturing **regional variability**
- ❑ First global model of **post-peak bleaching decay**
- ❑ New tools to **detect and correct survey timing bias**

Take-home message

Survey timing introduces a global, systematic downward bias in coral bleaching severity estimates